PHYSIO_FIELD_NOTES

Physiotherapist Field Interview — Analysis

MC 4206 · Interview 1 · 19 Sep 2026 · practising physiotherapist, Sri Lanka *Clinical experience: NHSL (National Hospital Sri Lanka) and Navajeevana rehabilitation centre, Tangalle.*

> Status: **living document**. More detail on wrist-specific motion is expected from the same source. Add subsequent interviews as Interview 2, 3 … and never edit earlier entries — append and supersede, same rule as the decision log.

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0. Why this document exists

Everything in the thesis up to now rests on *published* clinical evidence. This is the first **primary field evidence** the project has. Two of the claims we were making from the literature are now independently confirmed by a practitioner describing their own daily practice — unprompted, and in one case more damningly than the literature does.

It also surfaces one genuine contradiction with our stated target population, which is better to find now than in the viva.

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1. THE BIG ONE — two mechanisms, one deficit

What he said

> “Among stroke patients, there are many people with wrist drop. However, that usually occurs in the very early stage… As time goes on — after about a year — their muscles become stiff. Once stiffness sets in, movement stops happening, and the limb feels locked/rigid.”

And, separately:

> “Immediately after a stroke occurs, it starts with a floppy/loose state; that’s called flaccidity. Spasticity comes later.”

> “For a recent stroke patient, we use a machine called TENS… On the other hand, for patients with long-standing, chronic conditions, we mostly use stretching exercises.”

Why it matters

Wrist drop — failure to extend the wrist — is the deficit our device addresses. But it has two entirely different mechanical causes depending on time since onset:

	Acute / early	Chronic (~1 year +)
Tone	**Hypotonia** — flaccid	**Hypertonia** — spastic
Extensors	Weak / no drive	Weak, *plus* opposed by flexor stiffness
Passive stiffness `K_h`	**Low**	**High**
Velocity-dependent reflex `b_s`	**≈ 0**	**Present**
Resting wrist posture	Dropped by gravity	Held in flexion by tone
Current standard care	TENS (nerve stimulation)	Manual stretching
Voluntary EMG	Possibly weak or absent	Reduced but usually present

The *functional* complaint is identical — “I cannot straighten my wrist.” The *mechanical* signature is opposite.

This is not a problem. It is the thesis.

The Modified Ashworth Scale **cannot separate these** — it returns one ordinal grade that conflates weakness with stiffness (Mahmoud et al. 2023, already cited in D-01). Our device returns three independent numbers from the same session:

- `K_h` — passive elastic stiffness
- `b_s`, `θ̇_th` — velocity-dependent reflex magnitude and threshold
- voluntary torque capacity (from the assistive mode)

One ordinal grade versus three continuous, separately-identified physical parameters. That is the clearest statement of the project’s clinical contribution yet written, and it came from the field rather than from us.

But it forces an honest correction to D-09

Our stated target (D-09) is *chronic* stroke survivors with MAS ≤ 2 and residual EMG. The interview shows that:

- The **assistive** objective is most relevant **early**, when movement is still trainable and the limb is compliant — but that population is in-ward, ethics is harder, and TENS is the incumbent.
- The **measurement** objective is most relevant **chronically**, where MAS is the tool in use and where stiffness dominates.

These are not necessarily the same patients. **The thesis must say so explicitly rather than implying one population serves both objectives.** Written up as D-15.

Future work this opens (state it, do not attempt it)

Objective tracking of the **flaccid → spastic transition**. No instrument currently quantifies when and how fast that conversion happens in an individual. It requires longitudinal data we cannot collect in nine weeks, but naming it as the natural extension shows the panel you understand the clinical timeline you are working in.

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2. He validated our core premise more strongly than the literature does

On the Modified Ashworth Scale

> "Nowadays we don't strictly look at the scale sheet while doing it; when a patient comes, we bend the arm, estimate the grading in our mind, and proceed. We don't strictly refer to it as much now."

This is a practitioner stating that in routine practice the MAS is applied **from memory, by feel, without reference to the protocol sheet.** The published inter-rater reliability studies measure MAS applied *carefully, under study conditions*. Real-world reliability is therefore bounded above by those figures, not described by them.

Use this. It is primary evidence, properly anonymised and with consent, and it is far more persuasive to a panel than another citation. It also pre-empts the obvious attack — "clinicians already have a scale for this" — with the answer: *they have a scale they do not read.*

On goniometry

> "Nowadays, goniometers aren't really used much in everyday hospital settings… we can gauge an idea just by visual inspection."

> "We re-measure with a goniometer every 3 to 4 weeks."

Range of motion — the other quantity clinicians formally track — is also **eyeballed between 3–4 week intervals.** Our device records ROM to 0.088° resolution automatically, on every trial, as a by-product of the measurement we are already taking. That is a second objective output for zero additional hardware.

The value proposition, now with a number attached

> "If a patient can raise their leg by 30 degrees today, the next day they might raise it by 35 degrees… we won't see a massive leap overnight."

So: change in a good responder is on the order of **5° per session**, and the formal re-measurement interval is **3–4 weeks**. A device that measures every session at a repeatability of ~1° would detect real change **an order of magnitude earlier** than the current practice of visual estimation punctuated by monthly goniometry.

That sentence is your abstract. But it is only true if you measure the repeatability — see §4, Test 9.

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3. What it confirms about decisions already made

Our decision	Confirmed by
Target bandwidth `f_n` = 2 Hz (D-04)	Therapeutic exercises are slow and repetitive — "moving an object back and forth", finger opposition. Voluntary content is well under 1 Hz.
Velocity-graded spasticity test at 10–240°/s (Test 4)	MAS is assessed at **one uncontrolled velocity** — "we bend the arm". Controlled velocity grading is precisely what the clinical method lacks.
AAN with five discrete `K_d` levels (D-04)	"We start with simple movements… after that, we gradually increase the intensity." Graded progression is already standard practice — our levels are the *quantified* version of a judgement the physio currently makes by feel.
EMG-based relaxation gate on passive trials	"Some might not tell the truth or might be confused… we talk to them gently before starting." Subjective compliance with "stay relaxed" is unreliable in this population. Our EMG-RMS gate is the objective check — now justified by field evidence, not just rigour.
Excluding pronation/supination	He lists it as a separate movement (external/internal rotation at the shoulder; forearm rotation distinct again). Scoping to FE + RUD is consistent with how the assessment is actually partitioned.
Mode A montage: Ch1 = FCR, Ch2 = **ECR** (D-13)	Wrist drop is an **extensor** failure. ECR is the muscle that fails. We are recording the weak muscle on Ch2 — exactly the signal the autoencoder exists to reconstruct. The montage choice is clinically correct by accident; make it deliberate in the write-up.

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4. What it changes — concrete actions

4.1 NEW: Test 9 — mount/remount repeatability ⚠ highest-value addition

Encoder resolution is 0.088°. That is **not** the number a clinician cares about. The number they care about is: *if you take the device off and put it back on, how much does the answer move?* That is what sets the minimum

detectable change, and it is dominated by strap tension, limb placement and soft-tissue seating — not by the encoder.

Protocol. Per subject: run Tests 1 and 2, remove the device completely, re-don it, repeat — three times, same session, same operator. Report the standard deviation of `J_h`, `B_h`, `K_h` across mounts, and the resulting 95 % minimum detectable change.

Why this is the most publishable single number in the project. Without it, "we measure wrist impedance" is unfalsifiable. With it, you can write: *"minimum detectable change in K_h = X N·m/rad, against a MAS granularity of one ordinal grade."* That comparison is the paper.

4.2 NEW: proposed Test 10 — joint position sense (low cost, high clinical value)

> "For joint position sense, we ask the patient to close their eyes while we move their arm into different positions… If they lack position sense, they won't realise their arm is bent."

The device can do this objectively with **no additional hardware**: occlude vision, drive the joint to a target angle, return to neutral, ask the subject to reproduce it actively (or report direction of movement). The error in degrees is a continuous proprioceptive score where the clinical test is currently binary.

Propose it in the thesis as demonstrated-feasible future work if time does not allow it. Do not add it to the critical path.

4.3 Mechanical requirement we had not stated: adjustable device neutral

In wrist drop the hand rests **in flexion**. `wrist_params.m` currently sets `P.theta0 = 0`, and the hard stops are symmetric ±45° about the *device's* neutral.

If a patient's resting posture is 20° flexed and the device neutral is anatomical zero, the usable extension range collapses from 45° to 25° — on the side that matters clinically.

Action: the hand plate mounting must allow the device neutral to be set to the patient's own resting posture before strapping, with the offset recorded. Add `theta_zero_offset_deg` to the trial header (already present — now it has a reason) and raise the requirement with the mechanical team.

4.4 Protocol additions — fatigue, pain, and safety

> "We check whether they have fatigue — that means feeling tired — we check that first."

> "If it's an acute case… they will definitely have pain."

Test 4 alone is 5 velocities × 3 repeats = **15 imposed stretches**, and the fast ones deliberately provoke the stretch reflex. We currently specify 30 s rest between repeats and nothing else.

Add to the protocol and to the ethics application: - Fatigue and pain rating (simple 0–10) **before, mid-session and after**, recorded in the header. - A stated maximum session duration and total stretch count. - Explicit stop rules: pain report, fatigue rise beyond a threshold, or servo temp > 60 °C. - Acute patients have

pain and swelling — if the study ever extends to them, a separate protocol is required, not a relaxed version of this one.

4.5 SOAP-aligned data capture

Restructure the trial metadata header (`data/CSV_HEADER_TEMPLATE.txt`) to mirror the **S-O-A-P** structure so that the record is legible to a clinician without translation. Fields currently missing and now required:

- **Time since stroke onset** (and acute / subacute / chronic classification) — *the single most important covariate given §1, and we do not currently record it*
- Side of hemiplegia, **and hand dominance** (a dominant-hand deficit is functionally different from a non-dominant one)
- Chief complaint, verbatim
- Comorbidities — DM / hypertension / dyslipidaemia, and medication adherence
- Family history of stroke
- Caregiver support level, and independence in personal activities
- Tone classification: hypotonic / normal / hypertonic, plus MAS grade
- Presence of muscle wasting, swelling, shoulder asymmetry (observation, Objective phase)
- Sensation: light touch, pin-prick, joint position sense — pass/fail per site

4.6 Session-frequency and setup-time requirement

> "Every other day or once every two days… If it's an admitted ward patient — like at NHSL — we go and conduct sessions three times a day."

Three sessions a day means **setup time is a hard constraint**, not a convenience. If donning the device and placing electrodes takes 20 minutes, it will not be used in a ward.

Action: measure and report don/doff time. The SENIAM landmark marking template (already in the BOM) moves from "good practice" to "requirement" — re-finding the same electrode site quickly is what makes repeat sessions viable. Also note honestly that disposable gel electrodes at 3 sessions/day is not sustainable outside a study; dry electrodes are the deployment answer, and that belongs in Limitations.

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5. Competitive context — who we are actually up against

> "We rarely use machines for stroke patients… For them, we primarily do full rehabilitation. So most interventions are done manually by hand."

This is a **threat, and it must be stated in the thesis.** Stroke rehabilitation in this setting is manual-therapy-dominant. Our device is not entering a vacuum; it is asking for time in a session that is currently all hands.

The honest positioning:

- **We do not replace the therapist's hands.** A hand is better at assisting than our device and always will be within this budget.
- **A hand cannot measure.** It cannot impose a controlled 30, 60 or 120°/s stretch, it cannot record the resisting torque, and it cannot return a number. That is the entire gap we fill.
- **Versus TENS** (the incumbent machine for acute cases): different mechanism entirely — electrical stimulation of the nerve, not mechanical assistance. Not a competitor.
- **Useful fact:** TENS is a machine they already accept and use at the wrist. Clinician acceptance of *a device at the wrist* is therefore not the barrier. Session time is.

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6. Where the device's numbers would actually be used

> "Short-term goals are framed for 2 to 3 weeks, after which we evaluate whether the goal was met using tools like goniometers for range of motion, or manual muscle testing for muscle strength."

The **2–3 week short-term goal review** is the natural insertion point for our output. It is the moment where the clinician already has to produce evidence of progress, and where they currently reach for a goniometer they otherwise do not use.

Frame the device as *the instrument for the short-term goal review*, not as a therapy device competing for session minutes. That framing costs nothing and makes adoption a 2–3 week event rather than a daily one.

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7. Questions for Interview 2

He offered to continue, and said more wrist-specific detail is coming. Ask these, in this order — the first one is worth all the others combined.

The spec-defining question 1. *"What is the smallest change in a number that would make you change a patient's treatment?"* — This defines our required resolution **clinically** rather than by what the encoder happens to deliver. It converts Test 9's repeatability figure from interesting to pass/fail.

Wrist specifics 2. In wrist drop, what is the typical resting flexion angle? (sets the neutral offset, §4.3) 3. What wrist ROM do you target in early rehab versus late? Actual degrees. 4. Do you assess the wrist separately from the fingers, or together? Does finger flexor spasticity usually dominate the wrist? (we exclude fingers — need to know the cost) 5. When you assess tone at the wrist, roughly how hard do you push? (bounds our torque range) 6. How fast do you move the limb for the MAS? The scale says "as fast as possible" — is that what happens in practice?

Adoption 7. What would make you trust a number produced by a machine? 8. What would stop you using it in a session — time, cleaning, setup, something else? 9. If the device gave you one number per session, which quantity would you want it to be?

Population 10. Of the wrist-drop patients you see, roughly what proportion are within 3 months of onset versus over a year? (sizes the two populations in §1)

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8. What to cite, and how

- Interview content is **primary field data**. Cite as an anonymised expert interview with the date, setting and the consent statement. Do not name the individual or the patients.
- **Do not overstate it.** One practitioner, one setting, no sampling frame. It is corroborating field evidence and a source of design requirements — not a survey, and not evidence of prevalence or of general practice. Say that in the limitations.
- Where the interview *contradicts* our earlier assumption (§1, D-09), the contradiction goes in the thesis. A project that changed a stated target population because a practitioner told it something inconvenient reads as competent. One that quietly adjusted the wording does not.

Rev A — 19 Sep 2026. Interview 2 pending.

INTERVIEW 2 — wrist-specific

20 Sep 2026 · same source · with two reference images (goniometry panel, upper-limb synergy patterns) and a demonstration video link

Denser than Interview 1, and more useful. It contains one statement that goes directly against the project's original premise, one that breaks a protocol we had already written, and several that hand us hard numbers we did not have.

Taken in order of how much they change what we do.

A. "EMG is not very necessary for wrist evaluations"

> "Even without EMG, it doesn't matter much; EMG is not very necessary for wrist evaluations. Unless out of curiosity you want to check something for research purposes, it's not needed."

Do not skip past this. The field expert we went to says the project's headline technology is not needed. It must be in the thesis, quoted, and answered.

The honest answer, and it holds. He is describing **evaluation** — what a physiotherapist needs in order to assess and treat. For that he is right: MMT plus a goniometer covers it, and EMG adds nothing to an

assessment.

Our EMG is not an assessment output. It is a **control input** — the signal that tells the device the patient is *trying* to move, so that assistance can be contingent on effort rather than constant. Those are different functions with different customers:

	Purpose	Needs EMG?
Mode 1 — measurement	Produce K_h, b_s, $\dot{\theta}_{th}$, ROM for the clinician	**No.** Confirmed by this interview.
Mode 2 — assistance	Detect intent so assistance is effort-contingent, not constant	**Yes.** This is the only place it earns its place.

Three consequences, all of which should be acted on:

1. **It independently confirms the repositioning.** Mode 1 is clinically complete without EMG. We already knew this; a practitioner has now said it unprompted. The measurement objective does not depend on the ML pipeline working at all.
2. **Never frame the contribution as "EMG-based assessment."** A clinician will answer exactly as he did. The framing is *intent-driven assistance* and *instrumented measurement* — two claims, neither of which is "EMG tells you about the wrist."
3. **An uncomfortable organisational fact the team should say out loud.** The ML work supports the *secondary* objective only. That is not an argument for dropping it — it is an argument for making sure the primary objective does not wait on it. The five-field interface already enforces that separation; this is the clinical reason it was the right call.

B. The MAS is two passes. Our Test 4 is fifteen. ⚠ protocol break

> "We do not perform any movement repetitively on the patient during assessment. When evaluating with the Ashworth Scale, we flex and extend **at most twice**; we do not exceed that. Performing it more than a couple of times serves no purpose. We allow a short rest before continuing."

Test 4 as written is 5 velocities × 3 repeats = **15 imposed stretches**, deliberately provoking the stretch reflex each time. The clinician does two.

This is not only a tolerability problem. It is a measurement error.

"More than twice serves no purpose" is practitioner language for a real mechanical phenomenon: **thixotropic softening / stress relaxation**. Repeated stretching progressively reduces the measured resistance of the same limb. If we run 15 passes, passes 10–15 are measuring a limb that our own earlier passes have conditioned. Velocity and repetition number become confounded, and since we sweep velocity in order, the confound is *systematic*, not noise.

He has, without meaning to, told us our protocol produces a biased result.

Test 4 is redesigned

	Old	New
Passes per velocity	3	**2**
Velocity order	sequential	**randomised, per subject**
Rest between	30 s	**≥ 60 s, stated and logged**
Patient protocol	same as healthy	**2 passes total at one velocity**

And the confound becomes a finding — new Test 4b

Healthy subjects and the surrogate can take as many passes as we like. So:

Test 4b — stretch-conditioning decay. Fixed velocity, 10 consecutive passes, measure the peak resistive torque on each. Fit the decay. Report the time constant and the percentage lost from pass 1 to pass 10, plus the recovery after a stated rest.

Why this is worth doing: - It **quantifies the very effect the clinician describes qualitatively** and which no clinical scale captures. - It **justifies the reduced patient protocol with our own data** rather than by appeal to convention. - It tells us how long a rest is actually needed between trials — a number our protocol currently guesses at. - It is cheap: one subject, one velocity, ten passes, ten minutes.

This is the single most valuable thing to come out of Interview 2.

C. Finger posture is an uncontrolled variable — and it is large ⚠ new requirement

> “All these muscles are interconnected. The muscles in the fingertips, palm, and dorsal hand all attach around the eight carpal bones… both the hand and forearm exert an influence on the wrist.”

He is describing the **extrinsic finger muscles** — FDS, FDP, EDC — which originate on the forearm and cross **both** the wrist and the fingers. Their tension therefore depends on finger posture, and that tension passes through the wrist.

Consequence: measured wrist stiffness `K_h` depends on where the fingers are. A fist and an open hand give materially different answers on the same wrist. We do not currently control or record finger posture anywhere in the protocol.

This is almost certainly a dominant contributor to poor mount-to-mount repeatability, and without fixing it, Test 9 will produce a bad number we cannot diagnose.

Action

- **Mechanical:** add a **finger posture constraint** to the hand plate — a moulded grip or a light finger splint holding a fixed, repeatable posture. Cheap: a 3D-printed handle or a thermoplastic splint blank.
- **Protocol:** define one standard finger posture for all impedance trials and state it (relaxed semi-flexed is conventional). **Record it in the header.**
- **Same for forearm rotation.** The synergy-pattern image shows forearm rotation differs between the flexor pattern (supination) and extensor pattern (pronation). We do not control pronation/supination, so the forearm trough must **fix and record** forearm rotation in a stated neutral. Otherwise `K_h` is not comparable between subjects.
- **New header fields:** `finger_posture`, `forearm_rotation_deg`.

D. “End-feel” — the thing they admit is subjective, and we can put a number on it

> “It relies heavily on tactile feedback (‘end-feel’) that the therapist must feel… It can vary slightly between physios — **what feels stiff to me might feel less stiff to a male physio with more physical strength.**”

That second sentence is the most quotable line in either interview. It is a practitioner stating a **mechanism** for MAS inter-rater unreliability: the grade depends on the examiner’s own strength. Our device applies a known torque at a known velocity; examiner strength leaves the measurement entirely.

And “end-feel” has a direct engineering translation. It is the *shape of the torque–angle curve near end of range*: a hard end-feel is `dτ/dθ` rising steeply, a soft end-feel is a gradual rise. **Test 2 already records that curve.**

New derived metric — no new test needed

From the Test 2 quasi-static ramp, compute: - **Terminal stiffness** `K_end` — the slope over the last 20 % of the tested range - **End-feel index** `K_end / K_mid` — how much the curve hardens toward the end - The angle at which `dτ/dθ` exceeds a stated threshold

That converts the quantity the clinician says is hardest to teach, and most variable between examiners, into three numbers from data we are collecting anyway.

E. Hard numbers — and a derived specification for Test 9

> “If our target in the first week is about **2 centimetres**, we increase it to **3 centimetres** the following week, then **4**.”

Early-phase short-term goals are set in **centimetres of hand lift**, progressing about **1 cm per week**.

Converting to our units (stated assumptions — the exact measurement landmark must be confirmed in Interview 3):

Measured at	Distance from wrist axis	2 cm lift	Weekly 1 cm increment
3rd MCP head	~8.5 cm	≈ 13.6°	**≈ 6.8°**
Fingertip	~18 cm	≈ 6.4°	**≈ 3.2°**

So the clinically meaningful weekly change is single-digit degrees — somewhere between about 3° and 7°.

This gives Test 9 a pass criterion without waiting for Interview 3

To detect a weekly change reliably we need a minimum detectable change comfortably inside it. With a conventional factor of ~3:

> **Target MDC ≤ 2°, ideally ≤ 1°.**

That is a *derived requirement from clinical practice*, not a number chosen to flatter the hardware. Encoder resolution (0.088°) has enormous headroom; the whole budget is spent on mount repeatability and finger/forearm posture control — which is exactly why §C matters.

Also note: late-phase goals are **full ROM within 2–3 weeks**, via stretching **3×/day, 10 reps × 3 sets** — 90 stretches a day, performed at home by an untrained family member, with no measurement of any kind. Adherence and dose are completely invisible to the clinician. That is a real gap, and a clean *future work* statement (home logging), not something to scope into nine weeks.

F. ±45° is not enough for late-phase work ⚠ mechanical decision needed

Normal wrist ROM (AAOS reference values, to be confirmed against the standards he is sending): flexion ~80°, extension ~70°, radial deviation ~20°, ulnar deviation ~30°.

Our mechanical hard stops are **±45°**.

Use	±45° sufficient?
Impedance measurement near neutral	**Yes**, comfortably
Early-phase assistance (2–4 cm targets ≈ 6–24°)	**Yes**
Late-phase stretching to full ROM	**No.** We cannot reach 80° flexion or 70° extension.

Recommendation: make the hard stops **adjustable**, with ±45° as the default for measurement and assistance, openable to ±60° for ROM work under supervision. Raise with the mechanical team; it is a matter of stop-post

positions, not a redesign.

If it cannot be changed, it is a stated limitation: *the device addresses measurement and early-phase assistance; it does not deliver terminal-range stretching.*

G. New Test 11 — goniometer concurrent validity ⚠ high value, low cost

The goniometry image shows the standard technique: axis over the wrist, stationary arm along the forearm, moving arm along the hand.

A goniometer measures the hand segment relative to the forearm segment. Our encoder measures the device's joint angle. These are not automatically the same number — they diverge with axis misalignment, strap slip and soft-tissue movement.

If our degrees are not the clinic's degrees, none of our ROM output is clinically interpretable, and the comparison in §E collapses.

Test 11. On each subject, measure ROM simultaneously with the device and with a goniometer, across the range, several repeats. Report **Bland–Altman** agreement: bias and limits of agreement.

Cost: one goniometer (**add to the BOM — under USD 10**) and about twenty minutes per subject. This is the test that licenses the sentence *"the device's angle output agrees with clinical goniometry to within ±X°"* — without which we cannot claim clinical interpretability at all.

H. We operate in the gravity-eliminated regime — and can do something clever with it

> "Exercises are initially started in a **gravity-eliminated position** — usually supported on a bed, the arm rested on a table… Only after that do we progress to performing exercises **against gravity**… at least two to three weeks per stage."

Our decision to work in the horizontal plane (killing `τ_grav`) was made for measurement cleanliness. It turns out to place the device **exactly in the gravity-eliminated stage** — the stage where assistance is most needed and where the arm is supported on a table anyway. Our bench setup *is* the clinical setup for early-phase therapy. That is a fortunate alignment and worth stating.

The idea this unlocks: virtual load progression

The against-gravity stage, and the TheraBand/dumbbell stage after it, are both **load progression**. He describes progressing the load “bit by bit” in steps set by TheraBand colour and dumbbell increments — coarse, discrete, and unrecorded.

Our controller can add a **virtual resistive torque** — including a virtual gravity term `m·g·l·sin(θ − θ_g)` — in the horizontal plane. That gives: - **Continuously gradable** resistance, with a resolution no TheraBand ladder can match - Resistance that is a **recorded number**, not a colour - A measurement of the **work actually done** each session

It costs one term in the controller and no hardware. It is a strong secondary capability and it maps onto a progression the clinic already runs. Propose it; do not let it onto the critical path before Gate 1.

I. Scope — he asked for more, and that is usable both ways

> “It shouldn’t just be for flexion and extension; there are other wrist movements like pronation and supination… Then there are ulnar deviation and radial deviation… If multiple movements can be facilitated rather than just one, that would be great.”

> “Since you are starting on a small scale, **even incorporating at least one distinct movement mechanism would be very good.**”

Both halves are useful:

- **The second sentence is cover.** A practitioner explicitly endorsing a narrow scope is the answer to a panel member who asks “only two degrees of freedom?” Quote it.
- **The first sentence ranks future work by clinical demand.** Pronation/supination is *requested*, not merely possible. Adding a third DOF in nine weeks is not feasible and we will not pretend otherwise — but “the highest-priority extension, per practitioner request” is a far stronger future-work justification than “it would be nice.”

The patient base is wider than stroke

> “When nerve roots get damaged — like the ulnar nerve or radial nerve — those movements are completely lost… GBS (Guillain-Barré Syndrome) also starts paralysing from the distal joints.”

Radial nerve palsy is the classic cause of wrist drop, and it presents with a *non-spastic* limb — a focal, mechanically clean deficit, often with good recovery prognosis. That is in several ways an **easier and cleaner population** for an assistive device than stroke, and the measurement mode applies unchanged.

Broaden the claim at no cost: the platform addresses wrist drop *from any cause*, with post-stroke hemiparesis as the primary study population. Peripheral nerve injury and GBS named as directly applicable. This materially widens the impact statement without adding a single hour of work.

One honest limitation he handed us

> "Patients with isolated wrist paralysis are rare; usually the entire upper limb is affected."

A forearm-grounded device assumes the patient can position and stabilise the forearm. If shoulder and elbow are also involved, they cannot — which is precisely why the forearm trough and table support are **not optional**, and why the gravity-eliminated, supported configuration is the right one. State the assumption; do not let a reviewer find it first.

J. Prior art he pointed at — and the comparator we must cite

> "It's basically a glove… that glove mainly focuses on retraining finger movements. But **it didn't have anything for the wrist**… some Canadian brand."

The gap is stated by a practitioner: commercial hand-rehabilitation robotics is crowded at the **fingers**, and he sees nothing for the **wrist**. That belongs in the introduction, attributed to field interview.

Two things to do:

1. **Identify the glove.** Robotic rehabilitation gloves of that description on the market include SaeboGlove, Syrebo and Pisces Healthcare's rehabilitation robotic glove. Ask him for the brand — a named device the patient actually owned, in Sri Lanka, is a better citation than a generic category.

2. **Cite the real comparator — Myomo's MyoPro.** This is the commercial device closest to what we are building: a **myoelectric (EMG-driven) powered elbow-wrist-hand orthosis** for stroke and nerve injury, with peer-reviewed clinical evaluation (Usability, functionality and efficacy of a custom myoelectric elbow-wrist-hand orthosis; Myoelectric arm orthosis in motor learning-based therapy).

   We must cite MyoPro, and we must not claim EMG-driven upper-limb orthotics as novel. It exists, it is FDA-listed, it is sold. Our distinct position is not the EMG — it is

   a. wrist-specific, (b) **instrumented for measurement**, which MyoPro is not marketed as doing, and (c) at a research-prototype cost rather than a commercial orthosis price. Confirm current pricing before quoting any figure.

Failing to cite MyoPro would be the single most likely way to lose credibility in a viva. Find it now, cite it properly, and position against it deliberately.

K. A transcription caution — do not propagate this into the thesis

One passage is internally contradictory, almost certainly a translation artifact:

> "For wrist flexion and extension, the muscles on the anterior side of the forearm — meaning the upper/dorsal side of the arm — are most useful."

Anterior and *dorsal* are opposite sides. Later in the same interview he states it correctly: *"wrist extension is mainly influenced by the forearm muscles, and flexion is influenced by the muscles on the inner side of the forearm."*

Do not cite the first passage. The standard anatomy, which our montage already follows, is:

Motion	Prime movers	Compartment
Flexion	FCR, FCU (+ palmaris longus)	Anterior / volar
Extension	ECRL, ECRB, ECU	Posterior / dorsal
Radial deviation	FCR **+** ECRL/ECRB	both, cooperating
Ulnar deviation	FCU **+** ECU	both, cooperating

Our D-13 montage (Mode A: FCR + ECR; Mode B: FCR + FCU) is anatomically correct per the table above. **Correction, 2026-09-25 (D-25):** "SENIAM-consistent" overstates it: SENIAM's published sensor-location guidance covers a specific list of ~27 muscles (biceps brachii, triceps brachii, abductor pollicis brevis, etc.) and does not include FCR, FCU, ECR or ECU. Cite a standard anatomy text (e.g. Gray's Anatomy, or a hand/wrist-specific text) for the prime-mover table above, not SENIAM. Cite the interview for *practice*, as before. Keep those two kinds of claim separate throughout; it is the difference between a field interview strengthening a thesis and weakening it.

L. Questions for Interview 3

Closing the numbers 1. When you set a "2 cm" target — 2 cm measured **where**? Fingertip, knuckles, or the whole hand? *(closes the conversion in §E)* 2. Roughly how many degrees of flexion is the resting wrist position? *(sets the device neutral offset; he confirmed it rests flexed but gave no number)* 3. Please send the standard normal ROM values you use — we will compare ours against exactly your reference, not a textbook's. 4. The **spec question from Interview 1, still unanswered:** what is the smallest change in a number that would make you change a patient's treatment?

Protocol 5. Between the two Ashworth passes, how long do you rest? And between assessing different movements? 6. Do you standardise finger position while assessing the wrist, or does the hand just fall where it falls? *(directly tests §C)* 7. Is the forearm pronated, supinated or neutral when you assess wrist flexion/extension?

Device 8. The glove's brand name, if the patient remembers it. 9. Have you seen or heard of the MyoPro, or any powered wrist device, in Sri Lanka? 10. If our device could give you **one** number per session, which would you want? 11. Would you use a device that measured but did not treat — or must it do both to be worth session time?

Rev B — 20 Sep 2026. Interviews 1 and 2 recorded. Interview 3 pending.